D15 - Deep Learning with PyTorch

This downloadable PDF faithfully packages the supplied Knowledge Library authoring brief. It is source material, not a claim that a complete course book has been authored.

Role: optional Knowledge Library resource. It does not gate the main learning path.

Scope: Tensors, autograd, modules, data loaders, training loops, mixed precision concepts, checkpoints, and debugging.

Required deliverables

- Original English source in Markdown and plain text.
- Polished downloadable PDF, normally 25-40 pages unless the subject needs a justified different length.
- Prerequisite map and links to CodeQuest core objective IDs.
- Concept explanations, worked examples, practical exercises, solutions or answer guidance, glossary, references, and accessible descriptions for visuals.
- At least one applied mini-project and one failure-analysis section.
- Technical QA report covering code execution, formula checks, citations, and known limitations.

Authoring prompt

Write a rigorous but readable advanced guide on "Deep Learning with PyTorch" for CodeQuest AI Academy. Assume the stated prerequisites rather than repeating the entire core curriculum. Focus on depth, implementation judgment, tradeoffs, debugging, common failure modes, and professional practice. Use original language and examples. Do not reproduce copyrighted documentation or textbooks. Any code must run in the declared environment and include tests where appropriate. Any mathematical result must be checked. Produce matching Markdown, text, and PDF versions, then validate rendering and accessibility.